Carlos Romero Madrid

Computational Physicist · HPC Scientific Software Engineer · Creative Technologist

To: Fernando Cucchietti & Selection Committee
Data Analytics and Visualization Group (DAV)
Barcelona Supercomputing Center (BSC-CNS)
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I. GENESIS · SENSORY DATA 01

01. Career & Journey

Most applications start with a list of buzzwords. Let me begin with a tangible sensation: the rhythmic low frequency hum of **MareNostrum 5** calculating climate digital twins across thousands of GPUs, or the microtonal chime of an embedded motion sphere when my cat **Pipa** bats it across the room.

Data is not merely an abstract metric confined to spreadsheet rows and logs. Following pioneers like **Domestic Data Streamers**, I treat data as raw, physical matter—something to be felt, heard, and spatially inhabited.

Scientific computing is uncovering the deepest mechanics of our climate and energy systems. But when breakthroughs remain locked inside cold cluster nodes, *we have only finished half the mission*. My obsession is the **democratization of science**: transforming petascale simulations into unforgettable, visceral human experiences.

II. RIGOR & EXTREME COMPUTE 02

02. Education & Skills

My creative installations are grounded in first-principles theoretical and high-performance computing rigor:

- **Academic Core:** Double BSc in **Physics & Mathematics** (Univ. Seville, *3rd of Class*) + MSc in **Plasma & Fusion Physics** (Aix-Marseille Univ., *1st of Class*).

● POSICIÓN ACTUAL · CURRENT POSITION
BSC Performance Team (05/2025 – Present): Ocean-atmosphere coupling in **DestinationE** and **ClimateDT**, **ECLand** climate model development, and bottleneck analysis on **MareNostrum 5 (MN5)** & LUMI.

- **Fusion Team & CENER:** Parallelizing multi-physics simulation codes (ALYA-MAGNET, PETSc, oneAPI, OpenACC) and engineering real-time predictive wind plant data pipelines.

I pair continuously with **LLMs** as cognitive accelerators, moving seamlessly from C++/Fortran parallel kernels to browser audio synthesis and kinetic visual design.

III. STUDIO & PROTOTYPES 03

03. Creative Computing

MNusic — Real-Time Telemetry Sonification
Translating live GPU compute bursts, interconnect latency, and cooling thermal gradients into real-time harmonic soundscapes (Web Audio API) and generative particles. Making Europe's supercomputers audible.

Pipa-Music-Ball — Bio-Kinetic Interactive Sphere
Embedded ESP32 sphere with 6-DoF inertial sensors mapping feline kinetic velocity into microtonal scales. Proof that physical data can be playful, tactile, and non-anthropocentric.

Expo-Curator Studio (github.com/carrommad/expo-curator)
Open-source application for automated image classification, booklet publishing, and spatial gallery wall staging for exhibition curators and science dissemination.

Lens & Motion — Photography & After Effects Lab
Architectural light framing paired with procedural particle simulations and kinetic typography in Adobe After Effects for broadcast-grade visual essays.

IV. CONVERGENCE · FUTURE ACTION 04

04. Cover Letter to DAV Group

Having tracked your publications and exhibition work, I see an immediate symbiosis with DAV's core research pillars:

`Scientific Storytelling` `Interactive Viz` `Industrial AI & Digital Twins` `Creative Intelligence` `Urban Data Science`

Whether engineering urban digital twin visuals for **GDU-HPC**, public environmental resilience narratives for **AIR**, or active climate mobility for **REALLOCATE**, I combine deep HPC infrastructure roots with daring sensorial storytelling.

I want to join your team to co-create the most impactful scientific visualization installations in Europe.

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carrommad@gmail.com · carlos.romero@bsc.es
+34 611 423 133 · Barcelona, Spain